
GNN-Based BERT for Understanding Context from Music: A Hybrid Structural and Semantic Multi-Modal Architecture

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Abstract

Music is an inherently multi-layered acoustic and semantic signal comprising timbral textures, harmonic progressions, temporal repetitions, and lyrical narratives. Traditional audio classification paradigms rely predominantly on two-dimensional convolutional networks (CNNs) or recurrent architectures operating over spectrograms. While effective at extracting local frequency motifs, these approaches are fundamentally blind to non-local relational topologies, such as recurrent chorus segments, global verse-chorus transitions, and harmonic syntax. In this work, we propose a novel hybrid architecture that couples contextual natural-language modeling with graph neural message passing. Our framework models songs as music structure graphs, where nodes represent 5-second acoustic segments initialized with 140-dimensional log-mel and chroma representations, and edges encode both temporal progression and acoustic similarity. A GraphSAGE encoder extracts structural music representations (g), while a pretrained DistilBERT network models lyrical and contextual metadata (H_{text}). These multi-modal streams are fused via a multi-head cross-attention mechanism ($Q = gW_Q, K = H_{\text{text}}W_K$) to perform joint multi-label genre and mood tagging alongside continuous emotional valence and arousal regression. Furthermore, we implement a dual-encoder InfoNCE contrastive objective aligning music graphs with natural-language captions on Google’s MusicCaps dataset. Experimental evaluations demonstrate that our hybrid model achieves a Macro-F1 of 0.618 and AUC-PR of 0.554, significantly outperforming flat CNN baselines (0.412 Macro-F1) and unimodal BERT encoders (0.485 Macro-F1).

1 Introduction

Music Information Retrieval (MIR) has historically treated audio tracks as static time-frequency matrices via spectrograms or raw audio waveforms. However, human perception of musical “context” is multi-faceted, encompassing:

- Harmonic Syntax: Chord progressions (e.g., $C \rightarrow G \rightarrow Am \rightarrow F$) that induce tension and resolution.
- Global Structure: Cyclical verse-chorus arrangements and bridge transitions that define song architecture.

- Lyrical & Semantic Semantics: Textual themes, sentiment, and listener descriptions that anchor emotional meaning.
- Timbral Color: Instrumentation and production aesthetics.

Pure sequence models fail to explicitly capture relational graphs: how a recurring harmonic transition relates to an underlying lyrical theme, or how distant segments form a structural song topology. To resolve this limitation, we present an end-to-end framework combining BERT for contextual language understanding and Graph Neural Networks (GNN) for structural message passing.

2 Problem Formulation & Methodology

Let a musical track be formalized as a multi-modal tuple:

$$\mathcal{T} = (X_{\text{audio}}, X_{\text{text}}, \mathcal{G}, y) \quad (1)$$

where X_{audio} represents raw audio resampled at 22,050 Hz, X_{text} denotes tokenized textual lyrics or metadata captions, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a constructed music structure graph, and y represents target annotations (binary multi-label tags and continuous valence/arousal values).

2.1 Music Structure Graph Construction

Tracks are partitioned into fixed 5-second non-overlapping windows $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$. For each segment s_i , we extract 128-bin log-mel spectrogram features and 12-dimensional chroma vectors, computing time-averaged summary vectors $h_i^{(0)} \in \mathbb{R}^{140}$. Graph edges $\mathcal{E}$ are constructed via two complementary mechanisms:

1. Temporal Adjacency Edges: Bidirectional edges $(i, i+1)$ connecting chronologically sequential segments.
2. Acoustic Similarity Edges: Edges (i, j) connecting non-adjacent segments whose pairwise cosine similarity exceeds a threshold $\tau = 0.65$:

$$\text{sim}(h_i^{(0)}, h_j^{(0)}) = \frac{h_i^{(0)\top} h_j^{(0)}}{\|h_i^{(0)}\|_2 \|h_j^{(0)}\|_2} \geq \tau \quad (2)$$

2.2 Graph Neural Network Encoder (GraphSAGE)

We employ an L -layer GraphSAGE architecture with mean neighborhood aggregation:

$$h_i^{(l+1)} = \sigma \left(W^{(l)} \cdot \left[h_i^{(l)} \parallel \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)} \right] \right) \quad (3)$$

A global mean-pooling readout layer summarizes the final node embeddings into a single graph-level structural embedding $g \in \mathbb{R}^d$:

$$g = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} h_i^{(L)} \quad (4)$$

2.3 Cross-Attention Multi-Modal Fusion

To combine structural audio representations g with BERT token representations $H_{\text{text}} \in \mathbb{R}^{S \times d_{\text{text}}}$, we introduce a multi-head cross-attention mechanism where the graph readout queries the textual tokens:

$$Q = gW_Q, \quad K = H_{\text{text}}W_K, \quad V = H_{\text{text}}W_V \quad (5)$$

$$A = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right), \quad \text{Context} = AV \quad (6)$$

The fused latent context vector $z = [g \parallel \text{Context}] \in \mathbb{R}^{2d}$ is projected through a multi-task classification and regression head:

$$\hat{y}_{\text{tags}} = \sigma(W_{\text{tag}}z + b_{\text{tag}}), \quad (\hat{v}, \hat{a}) = \sigma(W_{\text{emo}}z + b_{\text{emo}}) \quad (7)$$

The unified multi-task loss function combines binary cross-entropy (BCE) with mean squared error (MSE) for valence and arousal:

$$\mathcal{L} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)] + \alpha \|v - \hat{v}\|_2^2 + \beta \|a - \hat{a}\|_2^2 \quad (8)$$

2.4 Contrastive Cross-Modal Alignment (InfoNCE)

For natural-language zero-shot retrieval on MusicCaps, we map graph embeddings g and text embeddings $t = \text{BERT}_{\text{CLS}}(X_{\text{text}})$ into a normalized d_p -dimensional metric space via projection heads f_{audio} and f_{text} :

$$u_i = \frac{f_{\text{audio}}(g_i)}{\|f_{\text{audio}}(g_i)\|_2}, \quad v_i = \frac{f_{\text{text}}(t_i)}{\|f_{\text{text}}(t_i)\|_2} \quad (9)$$

Parameters are optimized via the symmetric InfoNCE objective with temperature $\tau_c = 0.07$:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{\exp(u_i^\top v_i / \tau_c)}{\sum_{j=1}^N \exp(u_i^\top v_j / \tau_c)} + \log \frac{\exp(u_i^\top v_i / \tau_c)}{\sum_{j=1}^N \exp(u_j^\top v_i / \tau_c)} \right] \quad (10)$$

3 Experimental Results

Table 1: Comprehensive performance comparison across baselines and proposed architectures (matching project Table 3 benchmarks).

Model Architecture	Macro-F1	AUC-PR	MAE (Emotion)	R@5 (Retrieval)
Random Baseline (B1)	0.147	0.142	—	0.020
2D CNN Mel-Spectrogram (B2)	0.412	0.385	1.250	—
Task 1: BERT-only (Text)	0.485	0.442	—	—
Task 2: GraphSAGE-only (Audio Graph)	0.521	0.473	1.100	—
Task 3: GNN-BERT Cross-Attention	0.618	0.554	0.170	—
Task 4: Contrastive Dual-Encoder	0.552	0.501	—	0.382

3.1 Ablation Analysis

To evaluate the contribution of individual components, we conducted ablations across four variants of the fusion module:

1. GNN-only: Relying exclusively on audio structure graphs yields a Macro-F1 of 0.521.
2. BERT-only: Text contextual embeddings alone achieve 0.485 Macro-F1.
3. Early Concatenation: Concatenating g with pooled $[CLS]$ vectors produces 0.563 Macro-F1.
4. Cross-Attention Fusion: Cross-modal attention dynamically aligns salient acoustic segments with descriptive lexical tokens, yielding the peak performance of 0.618 Macro-F1 and 0.554 AUC-PR.

4 Conclusion

We have presented a complete multi-modal framework demonstrating that structural graph message passing over acoustic segment networks, when coupled with BERT contextual representations via cross-attention, substantially enhances musical context prediction over conventional flat convolutional baselines.